

Competitor & Client Report Module — Developer Specification

Product: AI Digital Marketing Automation Tool **Module:** Quick Scan / Competitor & Client Report (PDF Snapshot) **Document type:** Feature & Functionality Spec for Developers **Version:** 1.0

1. Purpose

This module lets a user run a "quick scan" on their own domain plus one or more competitor domains, and instantly generate a clean, client-ready PDF report comparing traffic, domain authority, backlinks, paid/ads metrics, and social media metrics — side by side. The goal is speed and presentability: something a marketer/agency can generate in one click and send/share directly with a client, without manually building a report in a slide deck or spreadsheet.

This is a **standalone module**, independent from the Keyword Research module — it does not share scope, endpoints, or database tables with keyword research, though both may live in the same overall product.

2. Core Features

1. Quick Scan Input

- User enters their own domain + up to 3–5 competitor domains.
- One click → system pulls all metrics below and compiles a report.

2. Traffic Metrics (Organic + Paid)

- Estimated monthly organic traffic + trend
- Estimated paid/PPC traffic share
- Top traffic-driving countries
- Top organic keywords (brief list, not full research — that stays in the Keyword Research module)

3. Authority Metrics

- Domain Authority (DA) / Domain Rating (DR) — depends on provider used
- Trust Score / Spam Score if available

4. **Backlink Metrics**

- Total backlinks
- Referring domains count
- New vs lost backlinks (last 30/90 days)
- Top 5 referring domains

5. **Paid/Ads Metrics**

- Estimated paid search traffic
- Number of active ad keywords
- Estimated monthly ad spend (where provider supports it)
- Sample ad copy/creatives if available (some providers offer this)

6. **Social Media Metrics**

- Follower counts across platforms (Instagram, Facebook, LinkedIn, X, YouTube) — user's own accounts via official APIs; competitor public counts via a social-listening/scraping-safe provider
- Engagement rate (avg likes/comments per post) where accessible
- Posting frequency (last 30 days)

7. **Side-by-Side Comparison View**

- Table/scorecard: You vs Competitor 1 vs Competitor 2 vs Competitor 3, metric by metric, with simple up/down indicators.

8. **One-Click Branded PDF Export**

- Auto-generate a clean, client-ready PDF: cover page (client/brand logo placeholder), comparison tables, 2–3 auto-written AI summary insights ("Your DA is 15% lower than Competitor A, but you have 2x more referring domains gained in the last 30 days"), and a simple recommendations section.
- PDF should be white-label-ready (agency logo instead of tool's own branding, if the plan allows it).

9. **Shareable Link (optional, alongside PDF)**

- Generate a view-only hosted link as an alternative to sending the PDF file.

10. Report History

- Save past scans so users/agencies can re-download or compare against a previous snapshot (e.g., month-over-month client reporting).
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3. Functional Requirements

ID	Requirement
FR-1	Accept 1 primary domain + up to N competitor domains as input
FR-2	Fetch traffic, DA/DR, backlink, and paid-ads data from a third-party SEO data provider
FR-3	Fetch social media metrics via official platform APIs (for the user's own connected accounts) and a social-data provider (for competitors' public metrics)
FR-4	Normalize all fetched data into one internal schema regardless of source provider
FR-5	Generate AI-written summary insights from the normalized comparison data (LLM call)
FR-6	Render report as a formatted PDF (server-side PDF generation)
FR-7	Allow branding customization (logo, agency name, color accent) on Pro/Agency plans
FR-8	Cache provider data per domain for a TTL (e.g., 24–72 hours) to avoid re-paying for identical scans
FR-9	Store report history per user/project so past reports can be re-downloaded, not just regenerated

ID	Requirement
FR-10	Rate-limit scans per plan tier (this pulls from multiple paid data providers per scan, so it is cost-sensitive)

4. Suggested Architecture

[Frontend UI]

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[API Gateway / Backend]

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|---> [Report Service]

| |---> [SEO Data Provider API] --- traffic, DA/DR, backlinks, paid-ads estimate

| | (e.g., SEMrush / Ahrefs / SimilarWeb / DataForSEO — pick 1, or blend 2 for coverage)

| |---> [Social Platform APIs] --- Instagram Graph API, YouTube Data API, LinkedIn API, X API

| | (official APIs for the user's own connected accounts)

| |---> [Social/Competitor Data Provider] --- for competitor public social metrics

| | (e.g., a Social Blade-style provider, since competitors won't grant API access)

| |---> [Claude API] --- generates the written summary/insights section

| |---> [PDF Rendering Engine] --- e.g., Puppeteer/wkhtmltopdf (HTML→PDF) or a hosted PDF API

| |---> [Database] (Postgres) --- report snapshots + history

| |----> [Storage/CDN] --- stores generated PDF files for download/share links

Note for developer: No single provider covers traffic + DA + backlinks + paid-ads + social equally well. Realistically this module blends 2–3 data sources (e.g., SimilarWeb or SEMrush for traffic/paid, Ahrefs/Moz for DA+backlinks, a social API/provider for social). Budget for multiple provider subscriptions, and design the internal schema so the frontend/report template doesn't care which provider a number came from.

5. Recommended Tech Stack

- **Backend:** Node.js (Express/NestJS) or Python (FastAPI/Django)
 - **Database:** PostgreSQL + Redis (caching, rate limits)
 - **PDF generation:** Puppeteer/headless Chromium (HTML → PDF) — recommended over low-level PDF libraries for easier design iteration
 - **AI/LLM:** Claude API for the written summary/insights section
 - **SEO/traffic/backlink data providers (pick/blend):**
 - SimilarWeb API — strong for traffic estimates
 - SEMrush API — traffic, paid ads, keyword overlap
 - Ahrefs API — strong for DA/DR and backlinks
 - Moz API — DA alternative
 - DataForSEO — cost-effective, covers multiple metric types
 - **Social data:**
 - Official APIs (Instagram Graph API, YouTube Data API, LinkedIn API, X API) for the user's own connected accounts
 - A third-party social analytics provider for competitor public metrics
 - **Storage:** S3-compatible storage or CDN for generated PDF files
 - **Frontend:** React/Next.js
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6. Data Model (Suggested Schema)

Table: `competitor_reports` | Field | Type | Notes | |---|---|---| | id | UUID | Primary key | | user_id | UUID | | | primary_domain | string | | | competitor_domains | JSON | array of domains | | status | enum | pending / completed / failed | | pdf_file_url | string | link to generated PDF in storage | | created_at | timestamp | |

Table: `report_metrics` | Field | Type | Notes | |---|---|---| | id | UUID | Primary key | | report_id | UUID | FK to competitor_reports | | domain | string | which domain this row belongs to (primary

or a competitor) | | organic_traffic | integer | | | paid_traffic | integer | | | da_score | float | | |
backlinks_total | integer | | | referring_domains | integer | | | new_backlinks_30d | integer | | |
lost_backlinks_30d | integer | | | paid_keywords_count | integer | | | est_ad_spend | float |
nullable | | social_metrics | JSON | per-platform follower/engagement data | | fetched_at |
timestamp | |

Table: report_branding | Field | Type | Notes | |---|---|---| | id | UUID | | | user_id | UUID | | |
logo_url | string | | | agency_name | string | | | accent_color | string | hex code |

7. API Endpoints (Suggested)

POST /api/reports/scan -> body: {primary_domain, competitor_domains[]} -> creates a report job

GET /api/reports/:report_id -> returns report status + data (once ready)

GET /api/reports/:report_id/pdf -> returns/generates the downloadable PDF

GET /api/reports/history -> list of past reports for the user

POST /api/reports/:report_id/branding -> body: {logo_url, agency_name, accent_color}
(Pro/Agency plans)

GET /api/reports/:report_id/share-link -> generates a view-only hosted link

8. LLM Prompt Notes (for Claude API integration)

- **Summary insights:** Send the normalized comparison data (as structured JSON) to Claude and ask for 2–4 short, factual sentences highlighting the most notable gaps/advantages (e.g., biggest DA gap, fastest-growing backlink profile, highest paid spend). Enforce that it only references numbers present in the data — no speculation.
 - Keep output format strict (plain text or JSON with a fixed set of fields) so it drops cleanly into the PDF template.
 - Low temperature recommended for consistency across reports.
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9. Non-Functional Requirements

- **Performance:** A full scan (multiple providers + PDF render) should run as an async job with a status endpoint — do not block the request/response cycle.
 - **Caching:** Cache provider data per domain for 24–72 hours; don't re-fetch identical domain data across different reports within that window.
 - **Cost control:** This module hits multiple paid third-party APIs per scan — enforce per-plan scan quotas (e.g., Free = 1 scan/month, Pro = 20/month, Agency = higher/unlimited with fair-use cap).
 - **Security:** Store all provider API keys in a secrets manager. Generated PDFs and share links should be access-controlled (signed URLs / expiring links where possible).
 - **Branding:** White-label customization should be gated behind the appropriate plan tier.
 - **Error handling:** If one data provider fails (e.g., social data unavailable for a competitor), still generate the report with a "data unavailable" note for that metric rather than failing the whole scan.
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10. What the Developer Needs Before Starting

- ☐ Decide primary data provider(s) for traffic/DA/backlinks/paid data (SEMrush, Ahrefs, SimilarWeb, Moz, DataForSEO — budget and coverage tradeoffs differ)
 - ☐ Decide provider for competitor social metrics (competitors won't grant API access, so this needs a third-party social-data provider)
 - ☐ Confirm which social platforms are in scope for the user's own connected accounts (Instagram, Facebook, LinkedIn, X, YouTube) and set up official API app approvals for each
 - ☐ Choose HTML-to-PDF rendering approach (headless Chromium/Puppeteer vs a hosted PDF API)
 - ☐ Confirm white-label/branding requirements (logo upload, agency name, color) and which plan tier unlocks it
 - ☐ Decide max competitor count per scan (affects cost per report) and per-plan scan quotas
 - ☐ Decide whether report history/re-comparison (month-over-month) is in v1 scope or a later phase
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11. Suggested Build Phases

Phase 1 (MVP): Domain + competitor input → traffic + DA + backlinks fetch → basic PDF export (no branding, no social/paid yet) **Phase 2:** Add paid-ads metrics + social media metrics

Phase 3: AI-written summary insights + side-by-side scorecard polish **Phase 4:** White-label branding + shareable link + report history/re-comparison